MILITARY SPECIFICATION

# POWER SUPPLY, CONSTANT FREQUENCY, FOR CLOCKS

## 1. SCOPE

1.1 This specification covers a source of frequency regulated 60 cycle per second (cps) power for operating synchronous-motor driven electric clocks used in shipboard control spaces.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirement for
MIL-C-3965 - Capacitors, Fixed, Nonsolid Electrolytic (Tantalum, Foil and Sintered-Slug), General Specification for.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-C-23339 - Clock, Electric, Internally-Illuminated, Wall-mounted

STANDARDS

MILITARY

MIL-STD-701 - Preferred and Guidance Lists of Semiconductor Devices.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Preproduction sample. - Equipment furnished under this specification shall be a product that has been tested and successfully passed the preproduction inspection specified herein. Preproduction samples shall be equivalent in all respects to the production units which the contractor proposes to furnish under this specification. Prior to submission of any production units for quality conformance inspection, the contractor shall furnish one complete preproduction sample which has been fully tested by the contractor (see 4.2) to determine compliance with this specification and which has been accepted by the cognizant Government inspector. The contractor shall record all data obtained during such tests in such form as is consonant with good engineering practice; one copy of such record shall be presented to the cognizant Government inspector at the time of inspection and one copy of such record shall be included with the preproduction equipment at time of its submission for approval to the command or agency concerned.

3.2 Description. - The equipment furnished under this specification shall provide alternating current (ac) power at a closely regulated frequency for operation of up to six wall-mounted internally-illuminated electric clocks of the type specified in MIL-C-23339.

3.3 General features. - In addition to the requirements specified herein, the equipment shall be in accordance with the requirements of MIL-I-983. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.)

Enclosure - degree of (see 3.6.1)
Shock, vibration
Primary power supply circuits (switching and fusing required) (see 3.5.6)

FSC 6645

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Power supply tolerances (normal power supply 115 volts, 60 cycles, single phase, ac)
Safety (personnel hazard)
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection
Terminal boards, and terminals
Wiring
Electrical insulation
Dielectric strength and insulation resistance clearances
Airborne noise (other than low noise applications)
Drawings - general
Drawings - preliminary
Drawings - working
Drawings - manufacturing
Bill of materials
Interchangeability and standardization
Manuals (see 6.1)
Repair parts - designation and marking
Reports
Workmanship

3.3.1 Operation. - The equipment shall be designed for continuous operation with a minimum of attention from operating personnel. No perating adjustments shall be required.

3.4 Performance requirements. -

3.4.1 Frequency and stability. - The output frequency shall be 60 cps, ± 0.018 percent or less, corresponding to a timekeeping accuracy of plus or minus 15 seconds per day, combined accuracy and stability, under all conditions of power supply variation (see 3.5.1) load conditions, (see 3.4.3) and ambient environmental conditions (as specified by MIL-I-983 for interior installation in surface ship). Effects of aging shall be included in this accuracy requirement.

3.4.2 Output voltage. - The output voltage shall be nominally 120 volts (rms) at 50 percent of full rated load (see 3.4.3), and under conditions of normal supply voltage and frequency (see 3.5.1).

3.4.2.1 Voltage regulation. - The output voltage shall be within the range of 108 to 132 volts rms for all combined conditions of power supply variations (115 ± 6 volts rms, 60 ± 3 cps) (see 3.3) and load variations (0 to 60 volt-amperes (va)) (see 3.4.3).

3.4.2.2 Voltage waveform. - The output voltage waveform shall be sinusoidal, clipped sinusoidal or square.

3.4.3 Output power. - The power supply unit shall be capable of delivering at least 60 va (corresponding to a load of at least six clocks) at any power factor between unity and 0.7 lagging under all operational conditions. It shall operate satisfactorily at all reduced load conditions, including no-load.

3.5 Electrical features. -

3.5.1 Power supplies. - The unit shall be designed to operate solely from normal power supply, 121 volts to 109 volts (rms), 63 cps to 57 cps, total harmonic content of 5 percent or less. Input shall not be grounded.

3.5.2 Power conversion. - Fully electronic power conversion shall be employed. Mechanical switching devices and rotating components shall not be used.

3.5.3 Semiconductor devices. - Semiconductor devices shall be employed where necessary and shall conform to MIL-STD-701. Electron tubes shall not be used.

3.5.4 Reference frequency. - The basic reference frequency shall be derived from an internal electronic oscillator. Frequency control shall be by means of a piezoelectric crystal, or vibrating resonant element. Frequency dividers may be employed when appropriate.



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3.5.4.1 The design shall be such as to prohibit an incorrect frequency output in the event of reference oscillator failure (see 4.5.3).

3.5.5 Output isolation. - The output circuit of the equipment shall be isolated from the unit chassis and from the ship's hull by means of a transformer.

3.5.6 Power switching and fusing. - Primary power switching shall be accomplished at the unit. A suitable switch shall be provided, accessible without removal of the urit enclosure. Suitable fusing shall be provided on the equipment to (a) prevent damage resulting from an overload at the output and (b) prevent damage to the ship's wiring resulting from equipment malfunction. The fuses shall be replaceable without removal of the unit enclosure.

3.5.7 Electrolytic capacitors. - Electrolytic capacitors may be employed where feasible to conserve space and weight. Tantalum electrolytic capacitors are preferred. Tantalum capacitors shall conform to MIL-C-3965.

3.5.8 Transient protection. - The equipment must withstand input line transients of 900 volts peak, 1 millisecond duration without damage.

3.6 Mechanical features. -

3.6.1 Enclosure. - The unit enclosure shall be of the dripproof type defined by MIL-I-983, designed for three-point bulkhead mounting (2 feet at the top and 1 foot at the bottom). Cable entrances and painting of interior and exterior surfaces shall conform to the requirements of MIL-I-983.

3.6.1.1 Size. - The complete unit shall be as small and compact as is consistent with good engineering practice. Limiting dimensions of 18 inches wide, 12 inches high, 12 inches deep shall not be exceeded.

3.6.1.2 Cooling. - The unit shall be designed such that all necessary heat transfer may be accomplished by natural convection. Forced air cooling shall not be permitted.

3.6.2 Shock and vibration. - The unit will withstand shock testing in accordance with MIL-I-983 and shall meet the vibration requirements for non-vital equipment. Shock and vibration mounts shall not be employed.

3.6.3 Airborne noise. - Airborne noise shall conform to the requirements of MIL-I-983 for other than low noise applications.

3.6.4 Structureborne noise. - When specified (see 6.1), the structureborne noise shall comply with MIL-STD-740. The grade of the equipment will be specified by the command or agency concerned.

3.6.5 Construction. - Modular construction shall be used wherever feasible. Modules shall be kept as small as possible consistent with good engineering practice. Modules shall be easily removable, clearly identified and keyed as necessary to permit interchangeability between similar equipments yet allow for repair and replacement aboard naval ships or repair facilities ashore.

3.7 Onboard repair parts. - The following items, if incorporated in the equipment, shall be furnished as on board repair parts:

| Name | Quantity |
|---|---|
| Electrolytic capacitors | One complete set. |
| Electronic circuit plug-in units (except reference oscillator modules) | One complete set. |
| Cable connectors | One each type and size. |

3.8 Stock repair parts. - When specified (see 6.1) stock repair parts shall be furnished based on a list of parts recommended by the supplier and as approved and identified by the command or agency concerned.

3.9 Manuals. - Manuals shall be type II of MIL-M-15071 and shall be furnished in the quantity specified (see 6.1).



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. - A preproduction sample of the equipment shall be submitted to the preproduction inspection as specified in table I. (See 6.2). Tests shall be performed in the order listed.

Table I - Preproduction inspection.

| Inspection | Requirement paragraph | Test paragraph |
|---|---|---|
| Examination | MIL-I-983 and 3.4.1 herein | 4.5.2 |
| Transients | 3.5.8 | MIL-I-983 |
| Output voltage | 3 4.2 | 4.5.4 |
| Frequency and stability | 3.4.1 | 4.5.3 |
| Supply line voltage and frequency variation | 3.3 and 3.4.2.1 | MIL-I-983 |
| Radio frequency noise interference | 3.3 | MIL-I-983 |
| Airborne noise | 3.6.3 | MIL-I-983 |
| Structureborne noise (see 6.1) | 3.6.4 | MIL-STD-740 |
| Accelerated life | 3 3 | MIL-I-983 |
| Dielectric strength | 3.3 | MIL-I-983 |
| Insulation resistance | 3.3 | MIL-I-983 |
| Enclosure | 3.3 and 3.6.1 | MIL-I-983 |
| Vibration | 3.3 and 3.6.2 | MIL-I-983 |
| Shock | 3.3 and 3.6.2 | MIL-I-983 |

4.3 Quality conformance inspection. - The number of equipment specified in 4.3.1.1 and 4.3.1.2 shall be subjected to group A and group B examinations and tests as specified in table II. Tests shall be performed in the order listed.

Table II - Quality conformance inspection.

| Inspection | Requirement paragraph | Test paragraph |
|---|---|---|
| Group A | | |
| Examination | MIL-I-983 and 3 4.1 herein | 4.5.2 |
| Dielectric strength | 3.3 | MIL-I-983 |
| Insulation resistance | 3.3 | MIL-I-983 |
| Output voltage | 3.4.2 | 4.5 4 |
| Group B | | |
| Frequency and stability | 3.4.1 | 4.5.3 |

4.3.1 Sampling for quality conformance inspection. -

4.3.1.1 Group A examination and tests. - Each equipment shall be subjected to each group A examination and test listed in 4.3 with acceptance based on 4.3.1.3.

4.3.1.2 Group B test. - All equipments of the same design and manufactured at approximately the same time shall be considered a lot. From each lot, a sample of equipment shall be selected, comprising at least



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10 percent of the lot size. Each equipment so selected shall be subjected to the group B test specified in 4.3, with acceptance based on 4.3.1.4.

4.3.1.3 Acceptance - Group A examination and tests. - The results of each examination and test shall be compared with the specification requirements. Failure to conform to the specification requirements for any test shall be counted as a defect, and the equipment shall not be submitted for delivery.

4.3.1.4 Acceptance - Group B test. - The result of the test shall be compared to the specification requirements. In the event of failure of any equipment to conform to this specification for the group B test, all equipments in the lot shall be subjected to the same test and corrected as necessary before submission for delivery.

4.4 The attention of the contractor is invited to the applicable paragraphs of MIL-I-983 for information concerning the tests.

4.5 Test procedures. -

4.5.1 Loading for tests. - Unless otherwise specified, for all tests in which the equipment is to be energized and operating, a load shall be provided such that the equipment supplies at least 60 va at 0.7 lagging power factor. This load shall contain at least one synchronous-motor driven electric clock, preferably in conformance with MIL-C-23339 and known to be in good operating condition, for monitoring timekeeping accuracy The time reference used for comparison shall be accurate to 0.005 percent or better.

4.5.2 General examination. - The general examination shall be conducted in accordance with MIL-I-983. The operating test will consist of at least 24 hours of continuous operation at 100 percent of rated output (60 va at 0.7 lagging power factor) with normal supply voltage and frequency. During this test, the equipment shall be checked for compliance with 3.4.1 and 3.4.2.

4.5.3 Frequency and stability test. - The equipment shall be run continuously during the following tests:

(a) The line voltage shall be set to 121 volts rms ± 1 volt while the line frequency is varied between 57 and 63 cps at a rate not to exceed 1 cps per minute. A load of 100 percent of rated output at 0.7 lagging power factor shall be connected.
(b) Repeat test (a) but with a load of 5 va or less at 0.7 lagging power factor.
(c) Repeat test (a) but set the line voltage to 109 volts rms ± 1 volt.
(d) Repeat test (c) but with a load of 5 va or less at 0.7 lagging power factor.

During the sequence of tests, the output voltage, waveform and frequency shall be monitored continuously for compliance with 3.4.1, 3.4.2.1 and 3.4.2.2. Minor deviations from the specified "percent of total test time" will be permitted to facilitate test scheduling. At the conclusion of the test, the reference oscillator shall be disabled and the unit output checked for compliance with 3.5.4.1. Any output voltage within the frequency range of 15-400 cycles will be cause for rejection.

4.5.4 Output voltage test. - The equipment output voltage shall be checked for compliance with 3.4.2 and 3.4.2.1 in accordance with MIL-I-983 at loads of less than 5 va (no load), 30 va and 60 va. This test may be conducted simultaneously with the frequency accuracy and stability test if desired.

4.5.5 Accuracy during other tests. - During all tests in which the equipment is energized and operating, and "test times" are of sufficient duration to produce meaningful results, the timekeeping accuracy shall be monitored for compliance with 3.4.1.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the



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using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice The information shall include nomenclature, Federal stock number, contract or order number, contractor's name and destination

5.1.4 Onboard repair parts. - Onboard repair parts and stock repair parts shall be prepared for the level of shipment as specified (see 6.1) in accordance with method III of MIL-P-116.

5.1.5 Technical manuals. - Technical manuals shall be prepared for the level of shipment as specified (see 6.1) in accordance with MIL-M-15071.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and level of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Quantity required.
(c) Structureborne requirements (see 3 6 4 and 4 2 (table I)).
(d) Quantity of stock repair parts (see 3.8).
(e) Quantity of technical manuals (see 3.9).
(f) Level of shipment for repair parts (see 5.1.4).
(g) Level of preservation, packaging and packing if other than 5.1 (see 5.2).

6.2 Preproduction. - Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - SH
(Project 6645-N019Sh)



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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